

NCERT Solutions For Class 12 Computer Science

Chapter 11 – Data Communication

1. What is data communication? What are the main components of data communication?

Ans: Data communication refers to the exchange of data between two or more networked or connected devices. The main components of data communication are sender, receiver, communication medium (transmission media), the message, and rules or protocols for communication.

2. Which communication mode allows communication in both directions simultaneously?

Ans: Full-duplex communication mode enables communication in both directions simultaneously.

3. Among LAN, MAN, and WAN, which has the highest speed and which one can cover the largest area?

Ans: LAN (Local Area Network) offers the highest speed, while WAN (Wide Area Network) covers the largest area.

4. What are three categories of wired media? Explain them.

Ans:

(a) Twisted Pair Cable: Consists of two insulated copper wires twisted together to minimize interference. Used in telephone lines and LANs. Types: Unshielded (UTP) and Shielded (STP).

(b) Coaxial Cable: Has a central copper core insulated and covered by an outer conductor (copper mesh) and a plastic cover. Offers better shielding and is used for higher frequency signals over longer distances.

(c) Optical Fiber: Carries data as light within a glass fiber, using refraction. High bandwidth, light-weight, immune to electromagnetic interference, but expensive and usually unidirectional (two cables needed for full-duplex).

5. Compare wired and wireless media.

Ans:

Wired Media (Guided)	Wireless Media (Unguided)
Physical cables (twisted pair, coaxial, fiber-optic) provide the path for data signals. More secure and stable but less flexible for mobility.	Uses electromagnetic waves through air (radio, microwave, infrared). Suitable for mobile and remote access but more prone to interference.
Installation and expansion may be costly.	Easy to set up and expand, especially in large areas.

6. Which transmission media carries signals in the form of light?

Ans: Optical Fiber transmission media carries signals in the form of light.

7. List out the advantages and disadvantages of optical fiber cable.

Ans:

Advantages:

1. High bandwidth and faster data transfer rates.
2. Light-weight and immune to electromagnetic interference.
3. Signals can travel long distances without much loss.

Disadvantages:

1. Expensive compared to other cables.
2. Unidirectional—needs two cables for full-duplex communication.
3. Requires specialized installation and maintenance.

8. What is the range of frequency for radio waves?

Ans: The range of frequency for radio waves is between 3 KHz to 1 GHz.

9. 18 Gbps is equal to how many bits per second?

Ans:

1 Gbps = 1,000,000,000 bits per second.

So, 18 Gbps = $18 \times 1,000,000,000 = 18,000,000,000$ bits per second.

10. HTTP stands for?

Ans: HTTP stands for HyperText Transfer Protocol.

11. Write short note on the following:

Ans:

a) HTTP

HTTP (HyperText Transfer Protocol) is the primary protocol used for accessing the World Wide Web. It follows a client-server model where requests are sent by clients (browsers) and responses are given by servers. HTTP defines the way information is formatted and transferred online.

b) Bandwidth

Bandwidth is the range of frequencies available for transmitting data through a channel. It is measured in Hertz (Hz). Higher bandwidth allows more data to be transferred per second.

c) Bluetooth

Bluetooth is a short-range wireless technology used to connect electronic devices (like phones, headphones, keyboards, printers) within about 10 meters at 1–2 Mbps. It creates a personal area network and is commonly used for wireless data exchange.

d) DNS

DNS (Domain Name System) is the naming system of the internet. It maps domain names (like google.com) to their

corresponding IP addresses, making web navigation easier by allowing users to use readable names instead of numbers.

e) Data transfer rate

Data transfer rate, or bit rate, is the number of bits transmitted between source and destination per second. It is measured in bits per second (bps). Higher transfer rates allow faster data communication.

12. What is protocol in data communication? Explain with an example.

Ans: In data communication, a protocol is a set of standard rules that the communicating parties (sender, receiver, and intermediates) must follow for successful and reliable data exchange.

Example: In the TCP/IP protocol, TCP handles reliable delivery and sequencing of packets, while IP manages addressing and routing. For instance, when sending an email, TCP ensures proper packet sequence and delivery, and IP routes the packets to the correct destination.

13. A composite signal contains frequencies between 500 MHz and 1 GHz. What is the bandwidth of a signal?

Ans:

Bandwidth = Maximum frequency – Minimum frequency

$$= 1 \text{ GHz} - 500 \text{ MHz} = 1,000,000,000 \text{ Hz} - 500,000,000 \text{ Hz} = 500,000,000 \text{ Hz} = 500 \text{ MHz}$$

Therefore, the bandwidth of the composite signal is 500 MHz.

